Externally Controlled Trial

- Case Study -

SuJin Heo, PhD
Project Leader, Real-World Operation, C&R RESEARCH

KSPM 2025 Medical Affairs Forum
2025-09-18

Agenda

- Introduction
- Definition of RWD/RWE
- FDA Guidance for Industry
- Externally Controlled Trial
- Case Study



Benefits from the use of RWD/RWE

- **Deloitte Center for Health Solutions (2022)**
  : Real-world evidence's evolution into a true end-to-end capability
  - 2022년에 글로벌 바이오제약 회사의 임원 17명을 대상으로 설문조사
  - 업계 그룹, 연구 기관 및 RWE 조직의 리더 10명을 인터뷰

[RWD/RWE의 이점]

- **2022** (설문조사 당시)
  - Disease progression, Incidence, Labeling revision (Safety)
  - Patient segmentation, Protocol feasibility (inclusion & exclusion)
- **2025** (2~3년 이후 기대하는 것)
  - Regulatory submission: *Synthetic,* ***External control arm***
  - ***Labeling revision (Effectiveness)***
  - 임상시험 디자인: 시간 & 비용 절감 (예: 기관 선정)
  - Tokenization: 서로 다른 데이터 소스를 익명으로 연결 (예: EHR, 청구 및 임상 시험 데이터)

허가 & 임상



[Reference] Real-world evidence use accelerates | Deloitte Insights
https://www.deloitte.com/us/en/insights/industry/health-care/real-world-evidence-study.html?icid=top_real-world-evidence-study

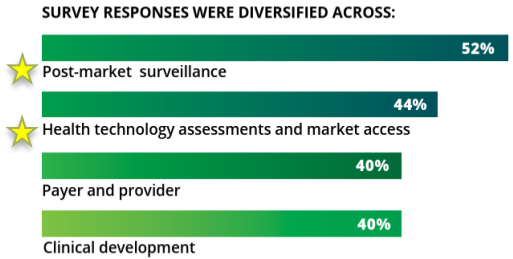
The diversification of RWE applications

- **Deloitte (2025):**

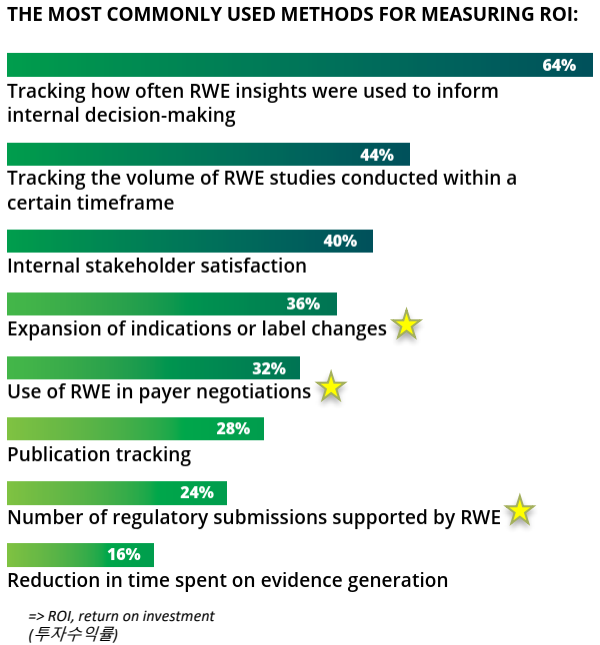
  Real-world evidence's evolution into a true end-to-end capability
  - 2024년 11월과 12월 사이 전 세계 25개의 생명공학 제약 회사의 경영진을 대상
  - 조직의 목표, 투자, 역량, 사용 사례 벤치마킹

- RWE의 사용
  - 내부 의사 결정 & 이해관계자 설득
  - **허가: 적응증 확장 & 라벨 변경**
  - 약가 협상

- RWE에서 가장 큰 가치를 보고 있는 영역
  - **시판 후 조사 (PMS, Post Marketing Surveillance)**
  - 약가
  - 임상시험 / Payer & Provider



허가 & 임상 & 약가



=> ROI, return on investment
(투자수익률)

[Reference] Converge™ by Deloitte | Real-world evidence and Generative AI: A powerful platform for innovation
https://www.deloitte.com/content/dam/assets-zone3/us/en/docs/services/consulting/2025/ada-compliant-2025-rwe-benchmarking-report.pdf?icid=top_/content/dam/assets-zone3/us/en/docs/services/consulting/2025/ada-compliant-2025-rwe-benchmarking-report.pdf

RWE: Neuroscience, 2019/01-2021/06

- **NDA/BLA Approvals : 3 Drugs**
- Indication
  - Rare disease
- Primary Endpoint (Efficacy)
  - Survival/Death
  - Functional
- RWD source
  - Database (registry)
  - Natural history

Table 1 Neuroscience-related applications from January 2019 to June 2021 adjudicated as containing RWE

Product	Indication	Key RWD efficacy endpoint	RWD source for control group
Onasemnogene abeparvovec[19]	Treatment of pediatric patients < 2 years of age with spinal muscular atrophy with bi-allelic mutations in the survival motor neuron 1 (SMN1) gene	• Survival at age 14-months defined by avoidance of the combined end point of either death or use of permanent ventilation (co-primary end point)	PNCR database, NeuroNEXT database
Risdiplam[20]	Treatment of spinal muscular atrophy	• Time to death or permanent ventilation (defined as ≥ 16 hour of noninvasive ventilation per day or intubation for > 21 consecutive days in the absence of, or following the resolution of, an acute reversible event or tracheostomy) from enrollment (secondary end point)	PNCR database, NeuroNEXT database
Viltolarsen[21]	Treatment of patients with DMD who have a confirmed mutation of the DMD gene that is amenable to exon 53 skipping	• Functional end points: TTSTAND, TTCLIMB, TTRW, 6MWT, NSAA, QMT (secondary end points)	CINRG natural history database

CINRG, Cooperative International Neuromuscular Research Group; DMD, Duchenne muscular Dystrophy; NCI, National Cancer Institute; NeuroNEXT, National Network for Excellence in Neuroscience Clinical Trials; NSAA, North Star Ambulatory Assessment; PFS, Progression Free Survival; PNCR, Pediatric Neuromuscular Clinical Research; POB, Pediatric Oncology Branch; QMT, Quantitative Muscle Testing; RCT, represents traditional randomized controlled trial; RWD, real-world data; RWE, real-world evidence; TTCLIMB, Time To Climb 4 Stairs; TTRW, Time To Run/Walk 10 Meters; TTSTAND, Time to Stand; 6MWT, Six-Minute Walk Test.

NDA, New drug application; BLA, Biologics license application;

[Reference] https://www.fda.gov/science-research/real-world-evidence/center-biologics-evaluation-and-research-center-drug-evaluation-and-research-real-world-evidence

Bloomfield-Clagett B, Rahman M, Smith K, Concato J. Use of Real-World Evidence in Neuroscience-Related New Drug and Biologics License Applications for Novel Therapeutics. *Clin Pharmacol Ther*. 2023;114(5):1002-1005. doi:10.1002/cpt.3018

RWE: FY 2020 – FY 2022*

NDA/BLA Approvals: 5

- 적응증
  - 희귀 질환
- 디자인
  - **비중재 연구** / 관찰연구
  - 외부대조군 연구
- 데이터
  - **레지스트리**
  - EMR

Table 1 Approved CDER product applications during FYs 2020–2022 with studies containing RWE that contributed to substantial evidence of effectiveness

Product	Date of approval	Approved indication		RWE study type	RWD source
Fosdenopterin	2/26/2021	Reduce the risk of mortality in patients with molybdenum cofactor deficiency (MoCD) Type A		Other[a]	Medical records
Tacrolimus	7/16/2021	Prophylaxis of organ rejection in adult and pediatric patients receiving allogeneic liver, kidney, heart, or lung transplants, in combination with other immunosuppressants		Non-interventional (observational) study: cohort	Registry
Vosoritide	11/19/2021	Increase linear growth in pediatric patients with achondroplasia with open epiphyses	(Study 1)	Externally controlled trial	Registry
			(Study 2)	Externally controlled trial	Registry
Abatacept	12/15/2021	Prophylaxis of acute graft versus host disease (aGVHD), in combination with a calcineurin inhibitor and methotrexate, in adults and pediatric patients 2 years of age and older undergoing hematopoietic stem cell transplantation (HSCT) from a matched or 1 allele-mismatched unrelated-donor.		Non-interventional (observational) study: cohort	Registry
Alpelisib	4/8/2022	Treatment of adult and pediatric patients 2 years of age and older with severe manifestations of PIK3CA-Related Overgrowth Spectrum (PROS) who require systemic therapy		Non-interventional (observational) study: cohort	Medical records

** FY 2020 – FY 2022: October 1, 2019–September 30, 2022*

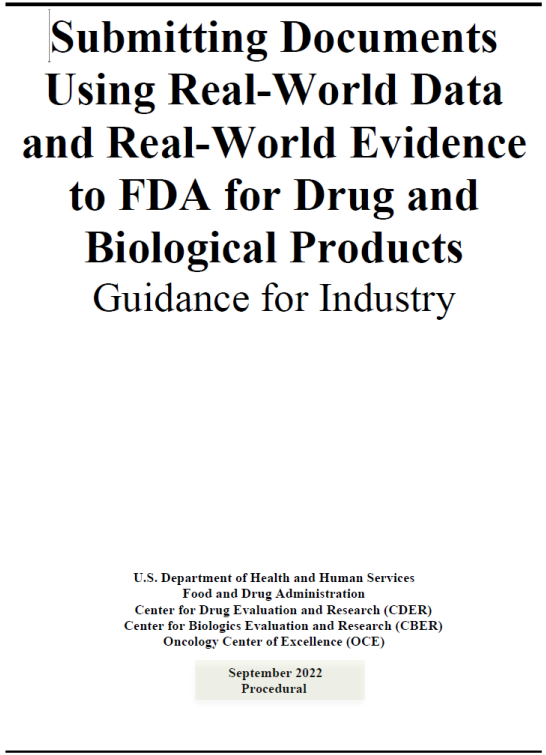
[a]Single-arm interventional trial with additional treatment data derived from RWD and comparator data derived from RWD.

[Reference]
Innes GK, Smith KA, Kuzucan A, et al. Real-World Evidence in New Drug and Biologics License Application Approvals During Fiscal Years 2020-2022. *Clin Pharmacol Ther*. 2025;118(1):85-89. doi:10.1002/cpt.3688

FDA: RWD – Definition (2022/09)

- **Real-world data (RWD)** are data relating to patient health status and/or the delivery of health care that are **routinely collected from a variety of sources**

- Examples include:
  - Electronic health record (EHR) data
  - Medical claims data
  - Product or disease registry data
  - Data obtained from digital health technologies
  - Data gathered from other sources that can inform on health status, such as questionnaires


Submitting Documents Using Real-World Data and Real-World Evidence to FDA for Drug and Biological Products

Guidance for Industry

U.S. Department of Health and Human Services
Food and Drug Administration
Center for Drug Evaluation and Research (CDER)
Center for Biologics Evaluation and Research (CBER)
Oncology Center of Excellence (OCE)

September 2022
Procedural

[Reference] https://www.fda.gov/regulatory-information/search-fda-guidance-documents/submitting-documents-using-real-world-data-and-real-world-evidence-fda-drug-and-biological-products

FDA: RWE - Definition

- **Real-world evidence (RWE)** is the **clinical evidence** about the **usage** and **potential benefits or risks** of a medical product derived from analysis of **RWD**

- **RWE Generation: Study design**
  - Interventional study: Randomized & controlled trials
    → Open label extension studies including RWD
  - Externally controlled trial: from RWD source
  - Non-interventional studies

TABLE 1 Perspective on when studies generate real-world evidence for studies of effectiveness and safety.

Study design	Is RWE generated? No	Yes
Interventional studies		
Randomized, controlled trials		
Real-world data (RWD) used to develop a study (e.g., to identify potential participants, select trial sites)	✓	
RWD used to assess impact of various enrollment criteria	✓	
Data from trial-provided digital health technology	✓	
Open-label extension studies not using RWD	✓	
★ RWD used for trial endpoint		✓
? Data from digital health technology used in non-research settings		✓
Open-label extension studies including RWD		✓
Externally controlled trials		
Single-arm trial with summary level estimate as comparator	✓	
External control arm data* from a clinical trial	✓	
★ External control arm data* from a RWD source		✓
Non-interventional studies		
Observational cohort study		✓
Case–control study		✓
Case-crossover study		✓
Self-controlled case series		✓

*Data on a specific group of patients as a comparator (e.g., not including publications with summary-level data).

[Reference]
1. https://www.fda.gov/regulatory-information/search-fda-guidance-documents/submitting-documents-using-real-world-data-and-real-world-evidence-fda-drug-and-biological-products
2. Rahman M, Dal Pan G, Stein P, et al. When can real-world data generate real-world evidence?. *Pharmacoepidemiol Drug Saf*. 2024;33(1):e5715. doi:10.1002/pds.5715

FDA: Regulatory Submission

- **Purposes of Using RWD/RWE**
  - Support safety and/or effectiveness for a product **not previously approved by FDA**
  - Support labeling changes for an **FDA-approved product**
  - Help to support or satisfy a **PMR/PMC**

신약 허가 & 라벨 변경 & 시판 후 의약품 연구

- ✓ Adding or modifying an indication
- ✓ Changing the dose, dose regimen, or route of administration
- ✓ Expanding the labeled indication of the product to a new population
- ✓ Adding comparative effectiveness information
- ✓ Adding or modifying safety information
- ✓ Proposing other labeling changes

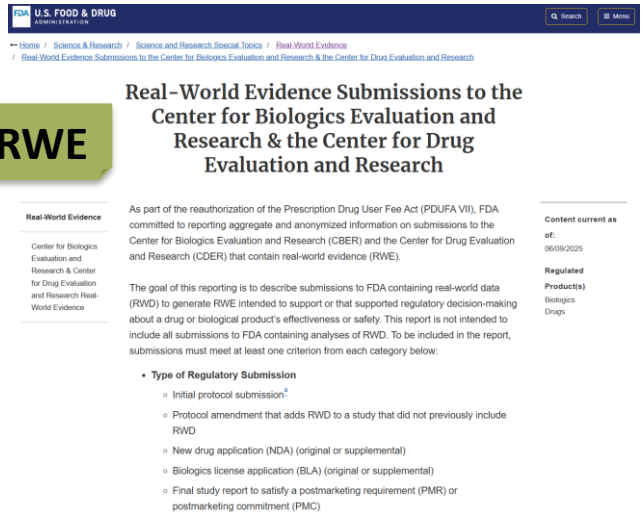
PMR, Postmarketing requirement; PMC, Postmarketing commitment

[Reference] https://www.fda.gov/regulatory-information/search-fda-guidance-documents/submitting-documents-using-real-world-data-and-real-world-evidence-fda-drug-and-biological-products

FDA: RWD & RWE

- FDA committed to reporting aggregate and **anonymized information** on submissions
- Columns will be added annually to represent submissions by **FY** from FYs 2023 through 2027
- As of **2025-06-10**

FDA RWE


U.S. FOOD & DRUG ADMINISTRATION

Home / Science & Research / Science and Research Special Topics / Real-World Evidence / Real-World Evidence Submissions to the Center for Biologics Evaluation and Research & the Center for Drug Evaluation and Research

Real-World Evidence Submissions to the Center for Biologics Evaluation and Research & the Center for Drug Evaluation and Research

Real-World Evidence

Center for Biologics Evaluation and Research & Center for Drug Evaluation and Research Real-World Evidence

As part of the reauthorization of the Prescription Drug User Fee Act (PDUFA VII), FDA committed to reporting aggregate and anonymized information on submissions to the Center for Biologics Evaluation and Research (CBER) and the Center for Drug Evaluation and Research (CDER) that contain real-world evidence (RWE).

The goal of this reporting is to describe submissions to FDA containing real-world data (RWD) to generate RWE intended to support or that supported regulatory decision-making about a drug or biological product's effectiveness or safety. This report is not intended to include all submissions to FDA containing analyses of RWD. To be included in the report, submissions must meet at least one criterion from each category below:

- **Type of Regulatory Submission**
  - Initial protocol submission[2]
  - Protocol amendment that adds RWD to a study that did not previously include RWD
  - New drug application (NDA) (original or supplemental)
  - Biologics license application (BLA) (original or supplemental)
  - Final study report to satisfy a postmarketing requirement (PMR) or postmarketing commitment (PMC)

Content current as of: 06/09/2025

Regulated Product(s): Biologics, Drugs

CDER


U.S. FOOD & DRUG ADMINISTRATION

Home / Science & Research / Science and Research Special Topics / Real-World Evidence / Real-World Evidence Submissions to the Center for Drug Evaluation and Research

Real-World Evidence Submissions to the Center for Drug Evaluation and Research

Real-World Evidence

Center for Biologics Evaluation and Research & Center for Drug Evaluation and Research Real-World Evidence

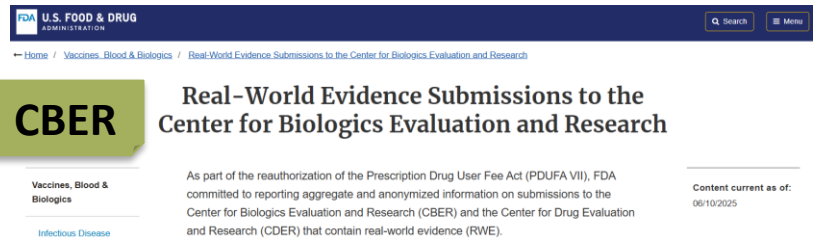
As part of the reauthorization of the Prescription Drug User Fee Act (PDUFA VII), FDA committed to reporting aggregate and anonymized information on submissions to the Center for Biologics Evaluation and Research (CBER) and the Center for Drug Evaluation and Research (CDER) that contain real-world evidence (RWE).

The tables below describe submissions to CDER containing RWE that meet reporting criteria. This report is not intended to include all submissions to CDER containing analyses of real-world data (RWD). Columns will be added annually to represent submissions by fiscal year (FY) from FYs 2023 through 2027.

Content current as of: 06/10/2025

Regulated Product(s): Drugs

CBER


U.S. FOOD & DRUG ADMINISTRATION

Home / Vaccines, Blood & Biologics / Real-World Evidence Submissions to the Center for Biologics Evaluation and Research

Real-World Evidence Submissions to the Center for Biologics Evaluation and Research

Vaccines, Blood & Biologics

Infectious Disease

As part of the reauthorization of the Prescription Drug User Fee Act (PDUFA VII), FDA committed to reporting aggregate and anonymized information on submissions to the Center for Biologics Evaluation and Research (CBER) and the Center for Drug Evaluation and Research (CDER) that contain real-world evidence (RWE).

Content current as of: 06/10/2025

[Reference] 1. https://www.fda.gov/science-research/real-world-evidence/real-world-evidence-submissions-center-biologics-evaluation-and-research-center-drug-evaluation-and
2. CBER: https://www.fda.gov/vaccines-blood-biologics/real-world-evidence-submissions-center-biologics-evaluation-and-research
3. CDER: https://www.fda.gov/science-research/real-world-evidence/real-world-evidence-submissions-center-drug-evaluation-and-research

RWE: FY 2023 – FY 2024

- NDA/BLA Approvals: 4 Drugs
  - Label expansion: 2 (Indication: 1, Loading dose: 1)
  - New drug approval: 2

	Approval Date	Drug Name	Company	[RWD] Data Source	[RWE] Study Design
CDER	2022-12-21	Actemra (tocilizumab)	Genentech	• **Electronic health records** (4) • Patient-level data from publications and national death records (2) • Product, disease, or other registry (2) • Medical claims	• **Non-interventional (observational) study** (4) • Externally controlled trial (3) • Randomized controlled trial (1)
CDER	2023-04-28	Vimpat (lacosamide)	UCB Inc.		
CDER	2024-02-13	Aurlumyn (iloprost)	Eicos Sciences, Inc.		
CBER	2024-03-18	**LENMELDY** (atidarsagene autotemcel)	Orchard Therapeutics (Europe) Limited		

[Reference]
1. CBER: https://www.fda.gov/vaccines-blood-biologics/real-world-evidence-submissions-center-biologics-evaluation-and-research
2. CDER: https://www.fda.gov/science-research/real-world-evidence/real-world-evidence-submissions-center-drug-evaluation-and-research

RWE: FY 2023 – FY 2024

[Label Extension]

1. **Tocilizumab**: New Indication - To treat hospitalized adult patients with COVID-19

   ✓ RWD: National death records to evaluate 28-day mortality

   (= Primary endpoint, randomized controlled clinical trial)

2. **Lacosamide**: New loading dose for pediatric patients

   ✓ RWD: Medical record data collated by the PEDSnet data network

[Reference]

RWE: FY 2023 – FY 2024

- BLA Approvals: 1 Drug
  - LENMELDY (Gene therapy)

- News release: CBER Director

  "*This is the **first** FDA-approved treatment option for **children** who have **this rare genetic disease***"

Rare Disease



U.S. FOOD & DRUG ADMINISTRATION

Search | Menu

Home / News & Events / FDA Newsroom / Press Announcements / FDA Approves First Gene Therapy for Children with Metachromatic Leukodystrophy

FDA NEWS RELEASE

FDA Approves First Gene Therapy for Children with Metachromatic Leukodystrophy

More Press Announcements

For Immediate Release: March 18, 2024

Español

Today, the U.S. Food and Drug Administration approved Lenmeldy (atidarsagene autotemcel), the first FDA-approved gene therapy indicated for the treatment of children with pre-symptomatic late infantile, pre-symptomatic early juvenile or early symptomatic early juvenile metachromatic leukodystrophy (MLD).

Metachromatic leukodystrophy is a debilitating, rare genetic disease affecting the brain and nervous system. It is caused by a deficiency of an enzyme called arylsulfatase A (ARSA), leading to a buildup of sulfatides (fatty substances) in the cells. This buildup causes damage to the central and peripheral nervous system, manifesting with loss of motor and cognitive function and early death. It is estimated that MLD affects one in every 40,000 individuals in the United States. There is no cure for MLD, and treatment typically focuses on supportive care and symptom management.

"This is the first FDA-approved treatment option for children who have this rare genetic disease," said Peter Marks, M.D., Ph.D., director of the FDA's Center for Biologics Evaluation and Research (CBER). "We remain committed to advancing scientific and regulatory principles that enable the efficient development and review of safe, effective and innovative products that have the potential to change patients' lives."

Content current as of: 03/18/2024

Regulated Product(s) Biologics

Follow FDA

Follow @US_FDA

Follow FDA

Follow @FDAMedia

[Reference]

1. CBER: https://www.fda.gov/vaccines-blood-biologics/real-world-evidence-submissions-center-biologics-evaluation-and-research
2. FDA New release: https://www.fda.gov/news-events/press-announcements/fda-approves-first-gene-therapy-children-metachromatic-leukodystrophy

RWE: LENMELDY

- BLA Approvals: 1 Drug

Approval Date	Drug Name	Company	[RWD] Data Source	[RWE] Study Design
2024-03-18	LENMELDY (atidarsagene autotemcel)	Orchard Therapeutics (Europe) Limited	• Medical claims • Product, disease, or other registry	• Externally controlled trial • Non-interventional (observational) study

- Indications and Usage:
  - LENMELDY is an

    autologous hematopoietic stem cell-based gene therapy

    indicated for the treatment of **children** with

    pre-symptomatic late infantile (PSLI),

    pre-symptomatic early juvenile (PSEJ)

    or early symptomatic early juvenile (ESEJ)

    metachromatic leukodystrophy (MLD)

Table 4: Demographics and Baseline Characteristics for Children with PSLI, PSEJ and ESEJ MLD Treated with LENMELDY (N=37)

Parameter	PSLI (N=20)	PSEJ (N=7)	ESEJ (N=10)
Age at Treatment (months) - Median (Min, Max)	12 (8, 19)	31 (11, 67)	70 (31, 140)
Male – n (%)	13 (65)	6 (86)	6 (60)
Race – n (%)	--	--	--
White/Caucasian	18 (90)	6 (86)	10 (100)
Asian	2 (10)	0	0
Black or African American	0	1 (14)	0
Ethnicity - Hispanic or Latino – n (%)	1 (5)	0	0

[Reference]
1. https://www.fda.gov/vaccines-blood-biologics/cellular-gene-therapy-products/lenmeldy
2. CBER: https://www.fda.gov/vaccines-blood-biologics/real-world-evidence-submissions-center-biologics-evaluation-and-research

RWE: LENMELDY (Clinical Review)

- **External Control**: Natural History Study

Comparator
: Untreated children
(Natural history)

6. DISCUSSION OF INDIVIDUAL STUDIES/CLINICAL TRIALS 24

6.1 Trial #1: A phase I/II clinical trial of hematopoietic stem cell gene therapy for the treatment of Metachromatic Leukodystrophy (OTL-200-201222) 24

6.2 Trial #2: A single-arm, open-label, clinical study of cryopreserved autologous CD34+ cells transduced with a lentiviral vector containing human ARSA cDNA OTL-200 for the treatment of early onset Metachromatic Leukodystrophy (MLD) 31

6.3 Expanded Access Program 34

6.4 External Control: Natural History Study in Subjects with Late Infantile and Early Juvenile Metachromatic Leukodystrophy (Study #OTL-200-204949). 34

6.4.1 Objectives 34

6.4.2 Design Overview 34

6.4.3 Population 34

6.4.4. Study Treatments or Agents Mandated by the Protocol 34

6.4.5. Directions for Use 34

6.4.6 Sites and Centers 34

6.4.7 Surveillance/Monitoring 34

6.4.8 Endpoints and Criteria for Study Success 35

6.4.9 Statistical Considerations & Statistical Analyses 35

6.4.10 Study Population and Disposition 35

7. INTEGRATED OVERVIEW OF EFFICACY 36

[Reference]
LENMELDY, BLA Clinical Review Memorandum: https://www.fda.gov/vaccines-blood-biologics/cellular-gene-therapy-products/lenmeldy

LENMELDY: External Control (Natural History)

- **Design**
  - **Observational, single-center**, study of the natural history of MLD that was initiated in **2004**
- Data Collection
  - **Prospective** assessments & **Retrospective** data collection

- ***Reviewer Comment:***
  - *This study, initiated in 2004, was not run concurrently with the clinical trials, which were initiated in 2010. Given that **no new therapies or changes to standard of care** for MLD occurred between 2004 and 2010, this was not considered a factor in the use as a comparator for the efficacy analysis. Limitations on the data obtained retrospectively are discussed below.*

[Reference]
LENMELDY, BLA Clinical Review Memorandum: https://www.fda.gov/vaccines-blood-biologics/cellular-gene-therapy-products/lenmeldy

LENMELDY: External Control (Natural History)

- Retrospective Data collection: **Medical chart review** & **Parent interviews**
  - Age, first nature of symptoms, achievement of early motor and language developmental milestones, GMFC-MLD classifications, brain MRI results

- ***Reviewer Comment:*** *An important limitation of this natural history study*
  - Use of retrospective data collection and **enrollment of patients after symptom onset**
  - Much of the data is collected through parental interviews, which are subject to recall bias
  - Incomplete data & Missing data in each individual subject
  - Additionally, retrospective data was collected at inconsistent timepoints for each subject

[Reference]
LENMELDY, BLA Clinical Review Memorandum: https://www.fda.gov/vaccines-blood-biologics/cellular-gene-therapy-products/lenmeldy

LENMELDY: External Control (Natural History)

- ***Reviewer Comment:*** *Significant limitations*
  - *The patients with EJ MLD are reported to have symptom onset between 24 and 75 months.* ***By definition****, EJ MLD may present as late as 7 years of age.*
  - *Additionally, there is significantly more phenotypic heterogeneity in EJ MLD. Therefore, the clinical review team identified concerns about the* ***suitability of the untreated patients*** *with EJ MLD*

Table 8: Characteristics of the LI and EJ Subjects in the Natural History Study

Parameter	LI (n=28)	EJ (n=17)
Sex, n (%)	-	-
Female	15 (54%)	9 (53%)
Male	13 (46%)	8 (47%)
Country of residence, n (%)	-	-
U.S.	1 (4%)	2 (12%)
Non-U.S.	27 (96%)	15 (88%)
Race, n (%)	-	-
White/Caucasian European	28 (100%)	17 (100%)
Asian	0	0
Black or African American	0	0
Age at symptom onset (months)[a]	-	-
Median	15	50
Minimum – maximum	9-28	24-75
Age at first assessment (months)	-	-
Median	19	60
Minimum – maximum	15-28	20-132
Age at enrollment (months)	-	-
Median	60	102
Minimum – maximum	31-168	55-215

Source: Reviewer analysis of Study #204949 ADSL Dataset
a In the event that a subject's exact age of onset is unknown, the earliest potential age is provided here.
Abbreviations: EJ, early juvenile; LI, late infantile ;U.S, United States

[Reference]
LENMELDY, BLA Clinical Review Memorandum: https://www.fda.gov/vaccines-blood-biologics/cellular-gene-therapy-products/lenmeldy

LENMELDY: External Control (Natural History)

- FDA concerns: **Interpretability** of the natural history GMFC-MLD scores

- Sponsor's FU Action

  1. **Questionnaire:** Reviewed by FDA

     ✓ "Clinician-Reported Outcome Evidence Dossier for the Gross Motor Function Classification for Metachromatic Leukodystrophy (GMFC-MLD)"

  2. Inter-rater reliability: Between prospective and retrospective GMFC-MLD assessments

     ✓ **Three blinded raters**: 86%, 89%, and 91%

[Reference]
LENMELDY, BLA Clinical Review Memorandum: https://www.fda.gov/vaccines-blood-biologics/cellular-gene-therapy-products/lenmeldy

LENMELDY: External Control (Natural History)

- FDA concerns: **Interpretability** of the natural history GMFC-MLD scores
- ***Reviewer Comment:*** *Significant limitations*
  - *Specifically, **one of the blinded raters** used to conduct the retrospective scoring was a study **investigator for the clinical trials**, which may make may bias assessments due to knowledge of the small number of subjects enrolled in both the natural history study and the treatment study.*
  - *During the pre-BLA meeting held with the Applicant on May 24, 2023 (CRMTS #14763), FDA advised the Applicant to re-run the study using qualified raters not involved in either the natural history or clinical trials. However, the **Applicant did not elect to re-run the study**.*
  - *Given these limitations and the other limitations in methodology described, the results of this dossier **did not inform the clinical review team's approach to the integrated efficacy analysis**.*

[Reference]
LENMELDY, BLA Clinical Review Memorandum: https://www.fda.gov/vaccines-blood-biologics/cellular-gene-therapy-products/lenmeldy

Guidance for Industry: RWE Submissions

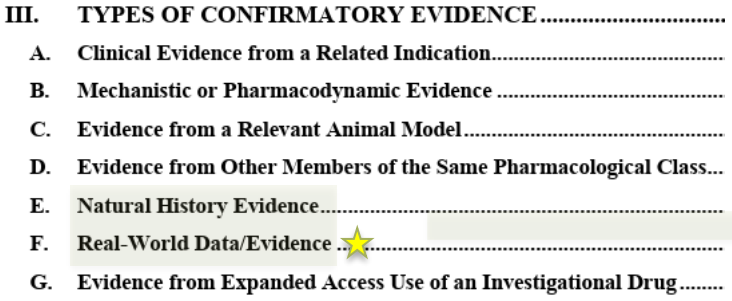
Draft Guidance

Category	Date	Selected Guidance
Regulatory Decision Making	August 2023	Considerations for the Use of Real-World Data and Real-World Evidence to Support Regulatory Decision-Making for Drug and Biological Products
Administrative	September 2022	Submitting Documents Utilizing Real-World Data and Real-World Evidence to FDA for Drugs and Biologics
	December 2023	Data Standards for Drug and Biological Production Submissions Containing Real-World Data
[RWD] Data Source	July 2018	Use of Electronic Health Record Data in Clinical Investigations
	December 2023	Real-World Data: Assessing **Registries** To Support Regulatory Decision-Making for Drug and Biological Products
	December 2023	**Digital Health Technologies** for Remote Data Acquisition in Clinical Investigations
	July 2024	**Real-World Data**: Assessing **Electronic Health Records** and **Medical Claims Data** To Support Regulatory Decision-Making for Drug and Biological Products
[RWE] Study Design	February 2023	Considerations for the Design and Conduct of **Externally Controlled Trials** for Drug and Biological Products
	March 2024	Considerations Regarding **Non-Interventional Studies** for Drug and Biological Products
	September 2024	**Integrating Randomized Controlled Trials** for Drug and Biological Products Into Routine Clinical Practice

[Reference]
https://www.fda.gov/science-research/science-and-research-special-topics/real-world-evidence
https://www.fda.gov/science-research/real-world-evidence/center-biologics-evaluation-and-research-center-drug-evaluation-and-research-real-world-evidence

Guidance for Industry: RWE Submissions

Category	Date	Related Guidance
Rare Disease ★	December 2023	**Rare Diseases**: Considerations for the Development of Drugs and Biological Products
	March 2019	Rare Diseases: **Natural History Studies** for Drug Development
Drug Approval	September 2023	Demonstrating **Substantial Evidence of Effectiveness** With **One Adequate and Well-Controlled Clinical Investigation** and **Confirmatory Evidence**
	October 2023	Risk-Benefit Assessment for New Drugs and Biological Products
Post-Marketing	January 2020	Long Term Follow-up After Administration of Human Gene Therapy Products
	August 2023	Postmarketing Approaches to Obtain Data on Populations Underrepresented in Clinical Trials for Drugs and Biological Products



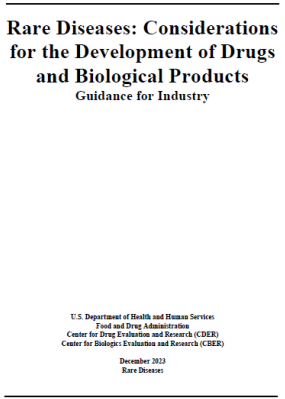
III. TYPES OF CONFIRMATORY EVIDENCE .....
A. Clinical Evidence from a Related Indication.....
B. Mechanistic or Pharmacodynamic Evidence .....
C. Evidence from a Relevant Animal Model.....
D. Evidence from Other Members of the Same Pharmacological Class...
E. Natural History Evidence.....
F. Real-World Data/Evidence .....
G. Evidence from Expanded Access Use of an Investigational Drug.....

[Confirmatory Evidence] RWD/RWE

- ✓ **Reliability** and **relevance** of the RWD source
- ✓ **Quality** of the study design
- ✓ Use of appropriate **prespecified** statistical **methods** and analyses

[Reference]
https://www.fda.gov/regulatory-information/search-fda-guidance-documents/demonstrating-substantial-evidence-effectiveness-one-adequate-and-well-controlled-clinical

Rare Disease: Drug Development (2023/12)


Rare Diseases: Considerations for the Development of Drugs and Biological Products
Guidance for Industry

U.S. Department of Health and Human Services
Food and Drug Administration
Center for Drug Evaluation and Research (CDER)
Center for Biologics Evaluation and Research (CBER)

December 2023
Rare Diseases

Background: Definition

- Rare disease or condition, in part, as a disease or condition that affects less than 200,000 persons in the United States

Consideration of Natural History Studies

- The natural history of rare diseases is often poorly understood, and the need for **prospectively** designed, **protocol-driven** natural history studies initiated in the earliest drug development planning stages cannot be overemphasized
- Robust natural history data can also help distinguish drug-related adverse events from underlying disease manifestations

[Reference]
https://www.fda.gov/regulatory-information/search-fda-guidance-documents/rare-diseases-considerations-development-drugs-and-biological-products

Rare Disease: External Control

- Background: For serious rare diseases with unmet medical need
- Definition
  - Outcomes in participants receiving the test treatment according to a protocol
    are **compared with** outcomes in a group of people **external to the clinical investigation**
    who had **not received the same treatment**
- External control designs are usually reserved for **specific circumstances**
  - Diseases with well-understood and characterized natural history
  - High and predictable mortality or progressive and predictable morbidity
  - Clinical investigations in which the **drug effect is large and self-evident**

[Reference]
https://www.fda.gov/regulatory-information/search-fda-guidance-documents/rare-diseases-considerations-development-drugs-and-biological-products

Rare Diseases: Natural History Study (19/03)

[Background]

- **Natural history study***:* Preplanned observational study intended to track the course of the disease
- Disease registries: A frequent platform to acquire the data for natural history studies
- Historical control: Potential use of natural history data as an external control in a clinical trial

[External Controls]

- **To be very similar** to the treated group in all respects (including disease severity, duration of illness, prior treatments, and any other aspects of the disease that could affect outcomes and the timing of outcomes)
  - **Nonconcurrent** vs *Concurrent*

[Concurrent External Control Arm]
However, in contrast with a completed natural history study, a concurrent control arm may not provide timely advice for planning the clinical trials

[Reference]
https://www.fda.gov/regulatory-information/search-fda-guidance-documents/rare-diseases-natural-history-studies-drug-development

FDA OCE: Oncology Studies using RWD

- Facilitating Review of RWD Studies
  - Oncology Quality, Characterization and Assessment of Real-World Data (QCARD) Initiative
  - Facilitating High-Quality Early RWD Study Proposals in Oncology

- Common Data Elements
  - RWD Source
  - Study Design
  - Statistical Plan
  - Data

Common Data Elements for Early Proposals of Oncology Studies Using RWD

Overall Study and RWD Source Information	Study Design Parameters	Statistical Plan	Data
• Study Sponsor • Real World Data Source Provider • Data Analyst(s) • Regulatory history of RWD proposal • Study Status (Planned, Ongoing) • Regulatory or Related Purpose(s) • Study Question(s)/ Objective(s) • RWD Source(s) • Including any Linked Data • Care Setting (as applicable) • Brief RWD Source Description • Rationale for RWD Source Selection	**Data Temporality** • Complete Original Source Data Years Available • Study Period • Data of Data Extraction (Analytic Dataset) • Initial Design Diagram (Graphical depiction of study design) **Population** • Disease or Condition for Intended Use • Indication Statement • Relevant Biomarkers • Study Eligibility Criteria • Expected Study Sample Size • External Validity: Representativeness of the source data to the intended use population (as applicable). **Exposure** • Type of Medical Product Exposure • Medical Product Exposure of Interest • Description of Exposure Ascertainment Method • Index Date Definition for Exposure Group **Comparator** • Comparator(s) of Interest • Description of the Comparator Ascertainment Method • Index Date • Definition for Comparator Group **Endpoints** • Primary Clinical Endpoint(s) Definition • Secondary Clinical Endpoint(s) Definition **Key Covariates** • Include any key covariates	Brief summary of the planned main statistical analyses of the primary endpoint(s)	• Source Data Availability and Auditing Plan • Data Provenance Documentation Plan • Data Quality Plan • Missing Data

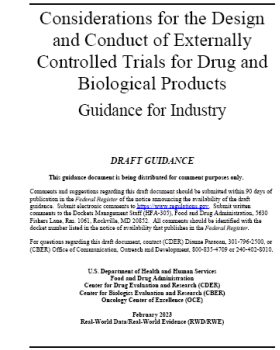
Externally Controlled Trials (ECT)

OCE, Oncology Center of Excellence

[Reference] https://www.fda.gov/news-events/fda-voices/facilitating-review-real-world-data-studies-oncology-qcard-initiative
https://www.fda.gov/about-fda/oncology-center-excellence/oncology-quality-characterization-and-assessment-real-world-data-qcard-initiative

ECT: Design (2023/02)

- **Baseline eligibility**: Inclusion/Exclusion criteria
  - Demography: Age, sex, race, socioeconomic status, geographic region
  - Disease: Severity, duration, specific signs and symptoms, performance status
- **Exposure definitions & windows**
  - Index date: Time zero
  - Drug formulation, dose, strength, route, timing, frequency, and duration
  - Dose modifications, interruptions, or discontinuations
  - Use of concomitant or supportive therapies (including non-prescription products)
- Well-defined and clinically meaningful **endpoints**
  - Tumor shrinkage: Blinded independent central review
- Minimize **missing data** & **bias** (Confounding factors)



Considerations for the Design and Conduct of Externally Controlled Trials for Drug and Biological Products

Guidance for Industry

DRAFT GUIDANCE

This guidance document is being distributed for comment purposes only.

Comments and suggestions regarding this draft document should be submitted within 90 days of publication in the *Federal Register* of the notice announcing the availability of the draft guidance. Submit electronic comments to https://www.regulations.gov. Submit written comments to the Dockets Management Staff (HFA-305), Food and Drug Administration, 5630 Fishers Lane, Rm. 1061, Rockville, MD 20852. All comments should be identified with the docket number listed in the notice of availability that publishes in the *Federal Register*.

For questions regarding this draft document, contact (CDER) Donna Pastoria, 301-796-2200, or (CBER) Office of Communication, Outreach and Development, 800-835-4709 or 240-402-8010.

U.S. Department of Health and Human Services
Food and Drug Administration
Center for Drug Evaluation and Research (CDER)
Center for Biologics Evaluation and Research (CBER)
Oncology Center of Excellence (OCE)

February 2023
Real-World Data/Real-World Evidence (RWD/RWE)

ECT: Comparator

[Reference] https://www.fda.gov/regulatory-information/search-fda-guidance-documents/considerations-design-and-conduct-externally-controlled-trials-drug-and-biological-products

ECT: Conduct & Analysis

- Data Source
  - **Clinical trials**
  - **RWD sources**
    - ✓ RWD sources *ay not capture the time of occurrence of an **intercurrent event** precluding accurate assessment of time-to-event endpoints such as progression-free survival*
    - ✓ *RWD sources that include information on alcohol intake collected using **structured questionnaires** are generally more reliable than patient-reported and clinician-documented values obtained during routine patient care*

- **Comparability** (Next page)

[Reference] https://www.fda.gov/regulatory-information/search-fda-guidance-documents/considerations-design-and-conduct-externally-controlled-trials-drug-and-biological-products

ECT: Comparability (1/3)

Focus of Comparison	Considerations for Data Comparability
1. Time periods	Various aspects of clinical care may change over time, such as the standard of care for the condition of interest, types of treatments, supportive care regimens, and **criteria for determining disease response or progression**. Such temporal differences are difficult to address using statistical analyses alone. It is important to consider whether and how different time frames in the treatment arm and the external control arm impact the interpretability of study findings.
2. Geographic region	Standards of care and other factors (e.g., access to care) that affect health-related outcomes can vary across geographic regions and health care systems. A balance of participants or patients across geographic regions and health care systems in an externally controlled trial, when possible, can help reduce the impact of confounding based on such differences.
3. Diagnosis	The criteria used to establish a diagnosis may differ based on practice variation or may have changed in the interval between when the treatment arm of the trial was conducted and when the data for the external control arm were collected. Sponsors should consider the diagnostic standards used and whether relevant clinical tests to establish a diagnosis were conducted and reported equally across the compared arms.

[Reference] https://www.fda.gov/regulatory-information/search-fda-guidance-documents/considerations-design-and-conduct-externally-controlled-trials-drug-and-biological-products

ECT: Comparability (2/3)

Focus of Comparison	Considerations for Data Comparability
4. Prognosis	Based on demographic and clinical characteristics—and if sufficient knowledge of relevant prognostic factors is available—prognostic indicators for the participants or patients in each arm of the trial should be evaluated and shown to be of sufficient similarity to permit an unbiased assessment of the treatment-outcome association.
5. Treatments	Attributes of the treatment of interest—including drug formulation, dose, route of administration, timing, frequency, and duration as well as specific rules for dose modifications, interruptions, discontinuations, and adherence—will have been prespecified or measured in the treatment arm. In contrast, specific aspects of a comparator treatment (as applicable) in the external control arm may not have been protocol-driven depending on the data source. Accordingly, sponsors should assess whether the external control arm data can be meaningfully compared to the treatment arm data.
6. Other treatment-related factors	Various treatment-related considerations, when relevant, include (1) previous treatments received (e.g., lines of therapy in patients with cancer), (2) medications received concomitantly that can affect the outcome of interest, or (3) predictive biomarkers (e.g., genomic testing) related to the treatment of interest. When differentially distributed across groups being compared, such factors can threaten an assessment of the drug-outcome association.

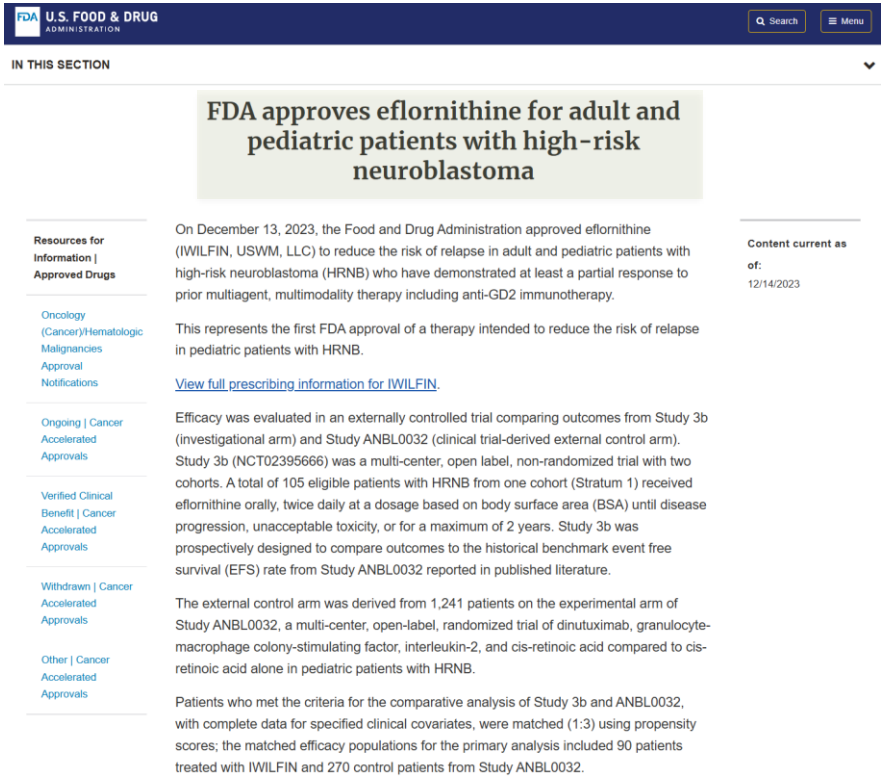
ECT: Comparability (3/3)

Focus of Comparison	Considerations for Data Comparability
7. Follow-up periods	Designation of the index date should be consistent between the treatment arm and the external control arm, and the duration of follow-up periods should be comparable across compared arms.
8. Intercurrent events	The relevance of intercurrent events across treatment arms should be assessed, including differential use of additional therapies after initiation of the treatment of interest.
9. Outcome	Whether endpoints used in an externally controlled trial can be reliably and consistently measured across the external control arm and the treatment arm will be influenced by several factors, including the definitions of the endpoints, the data source for the external control arm, and the potential for the outcome to be influenced by knowledge of treatment received. In addition, sponsors should be able to apply the same criteria for the evaluation and timing of outcome assessments across both arms of the externally controlled trial.
10. Missing data	The extent of missing data in the external control arm should be assessed before conducting an externally controlled trial to evaluate feasibility (when such data are available). When analyzing results from such a trial, the extent of missing data in both the treatment and external control arms should be assessed to examine the potential impact of missing data.

[Reference] https://www.fda.gov/regulatory-information/search-fda-guidance-documents/considerations-design-and-conduct-externally-controlled-trials-drug-and-biological-products

RWD/E: Case Study

Drug Name	**IWILFIN (eflornithine)**
Company	USWM, LLC
Approval Date	2023-12-13
Submission Classification	New Formulation or New Manufacturer
FDA Review	Priority, Orphan
RWE: Study Design	Externally controlled trial
RWD: Data Source	**Clinical trial-derived external control arm**
Comparative analysis	Propensity score matching (1:3) ⇒ Treated: 90 patients ⇒ Control: 270 patients



U.S. FOOD & DRUG ADMINISTRATION

Search | Menu

IN THIS SECTION

FDA approves eflornithine for adult and pediatric patients with high-risk neuroblastoma

Resources for Information | Approved Drugs

- Oncology (Cancer)/Hematologic Malignancies Approval Notifications
- Ongoing | Cancer Accelerated Approvals
- Verified Clinical Benefit | Cancer Accelerated Approvals
- Withdrawn | Cancer Accelerated Approvals
- Other | Cancer Accelerated Approvals

On December 13, 2023, the Food and Drug Administration approved eflornithine (IWILFIN, USWM, LLC) to reduce the risk of relapse in adult and pediatric patients with high-risk neuroblastoma (HRNB) who have demonstrated at least a partial response to prior multiagent, multimodality therapy including anti-GD2 immunotherapy.

This represents the first FDA approval of a therapy intended to reduce the risk of relapse in pediatric patients with HRNB.

View full prescribing information for IWILFIN.

Efficacy was evaluated in an externally controlled trial comparing outcomes from Study 3b (investigational arm) and Study ANBL0032 (clinical trial-derived external control arm). Study 3b (NCT02395666) was a multi-center, open label, non-randomized trial with two cohorts. A total of 105 eligible patients with HRNB from one cohort (Stratum 1) received eflornithine orally, twice daily at a dosage based on body surface area (BSA) until disease progression, unacceptable toxicity, or for a maximum of 2 years. Study 3b was prospectively designed to compare outcomes to the historical benchmark event free survival (EFS) rate from Study ANBL0032 reported in published literature.

The external control arm was derived from 1,241 patients on the experimental arm of Study ANBL0032, a multi-center, open-label, randomized trial of dinutuximab, granulocyte-macrophage colony-stimulating factor, interleukin-2, and cis-retinoic acid compared to cis-retinoic acid alone in pediatric patients with HRNB.

Patients who met the criteria for the comparative analysis of Study 3b and ANBL0032, with complete data for specified clinical covariates, were matched (1:3) using propensity scores; the matched efficacy populations for the primary analysis included 90 patients treated with IWILFIN and 270 control patients from Study ANBL0032.

Content current as of: 12/14/2023

[Reference] https://www.accessdata.fda.gov/scripts/cder/daf/index.cfm?event=overview.process&ApplNo=215500
https://www.fda.gov/drugs/resources-information-approved-drugs/fda-approves-eflornithine-adult-and-pediatric-patients-high-risk-neuroblastoma

FDA: Oncology RWE Program

- Case Study: Externally Controlled Trial
  - Presentation: FDA Oncology RWE Program Lead, EMA Workshop (2024-06-14)



Oncology RWE Program Lead



Donna R. Rivera, PharmD, MSc, FISPE is the Associate Director for Pharmacoepidemiology in the Oncology Center of Excellence at the U.S. Food and Drug Administration. She leads the Oncology Real World Evidence (RWE) Program, focused on the regulatory review of Real World Data (RWD) and RWE in oncology across CDER, CBER, and CDRH, management of the RWD research portfolio strategy, engagement and educational programs, and development of regulatory policy to support the OCE mission. As a pharmacist and pharmacoepidemiologist, Dr. Rivera has interests in applications of artificial intelligence in learning healthcare, pragmatic study designs and novel methodological approaches, and appropriate uses of RWD for drug development to increase access of effective therapies to patients.

[Reference]
https://www.fda.gov/about-fda/oncology-center-excellence/oncology-real-world-evidence-program
https://www.ema.europa.eu/en/documents/presentation/presentation-externally-controlled-trials-oncology-drivera-us-fda_en.pdf

IWILFIN (eflornithine)

[Review process]

1. **External control**

   : Appropriateness of Use

2. Assessment of

   Adequate and Well Controlled Study

3. Establishing **Substantial Evidence**

4. Overall Risk: Benefit Assessment

Eflornithine (DFMO)

Regular Approval on December 13, 2023

FDA

Indication	To reduce the risk of relapse in adult and pediatric patients with high-risk neuroblastoma who have demonstrated at least a partial response to prior multiagent, multimodality therapy including anti-GD2 immunotherapy.
Context	High unmet need in a rare, life-threatening disease
Clinical Evidence	★ Single-arm clinical trial with clinical trial data derived EC and confirmatory evidence **Investigational Arm:** single-arm trial (Study 3(b), Stratum 1) of 90 treated patients most of whom were previously enrolled on ANBL0032 **External Control Arm:** National Cancer Institute/Children's Oncology Group-sponsored clinical trial (Study ANBL0032) of 270 patients propensity-score matched to DFMO-treated patients on 11 demographic and clinical characteristics **Strong mechanistic non-clinical data**
Outcomes	Event-Free Survival HR=0.48 (95% CI: 0.27, 0.85) Overall Survival HR=0.32 (95% CI: 0.15, 0.70)

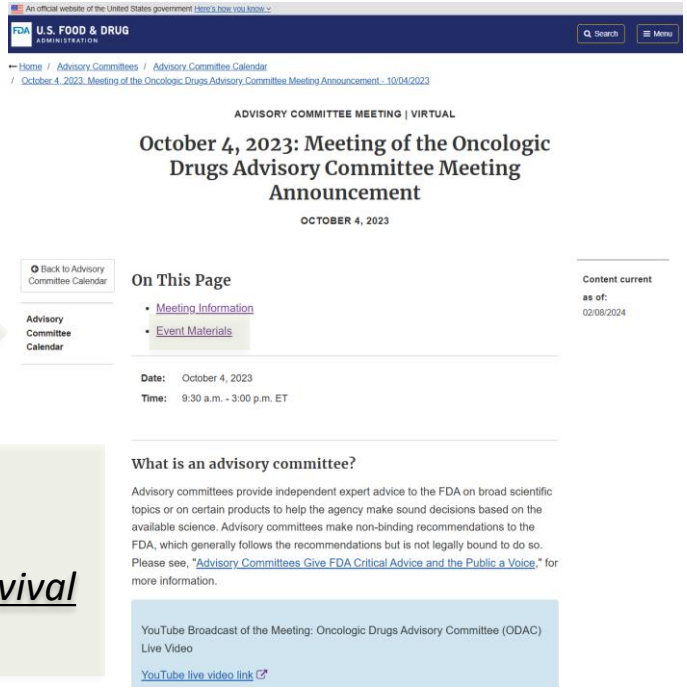
21

[Reference] https://www.ema.europa.eu/en/documents/presentation/presentation-externally-controlled-trials-oncology-drivera-us-fda_en.pdf

FDA: IWILFIN (eflornithine)

- ODAC: FDA's Oncologic Drugs Advisory Committee (2023-10-04)
- Virtual Meeting (Live): YouTube (6 hours)





[Voting Question]
*Has the Applicant provided **sufficient evidence** to conclude that DFMO improves event-free survival in patients with high-risk neuroblastoma?*

[Reference]
1. YouTube: https://www.youtube.com/watch?v=mavF5RvkJyk
2. Meeting Materials: https://www.fda.gov/advisory-committees/advisory-committee-calendar/october-4-2023-meeting-oncologic-drugs-advisory-committee-meeting-announcement-10042023#event-information

Indication

- Sponsor’s Presentation: US WorldMeds



[Reference] https://www.fda.gov/advisory-committees/advisory-committee-calendar/october-4-2023-meeting-oncologic-drugs-advisory-committee-meeting-announcement-10042023#event-information

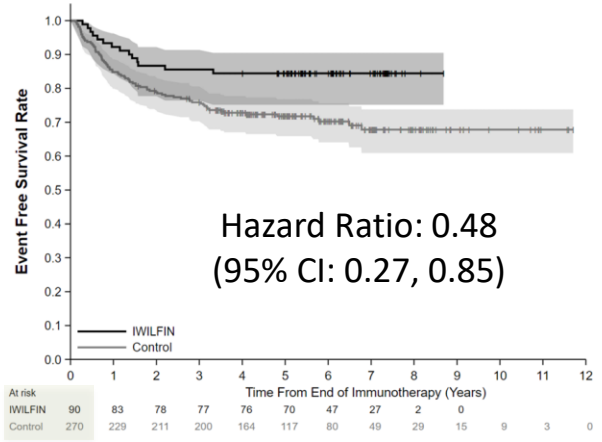
FDA Label: Clinical Studies

- Study 3b: **Investigational arm**
  - A multi-center, open label, non-randomized trial (NCT02395666)

- Study ANBL0032

  : **Clinical trial-derived external control arm**
  - 1,241 patients
  - A multi-center, open-label, randomized trial

Figure 1: Kaplan-Meier Curve for Event Free Survival for Protocol-Specified Primary Analysis in the Externally Controlled Trial



Propensity Score Matching (90 vs 270 patients)

[Reference] https://www.accessdata.fda.gov/drugsatfda_docs/label/2023/215500s000lbl.pdf

Propensity Score Matching

- Control Group **[N = 270]**

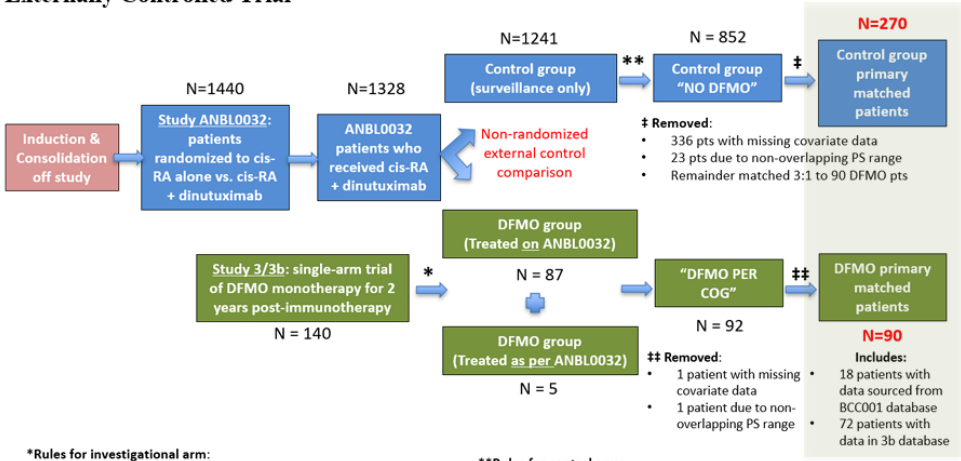
  : No DFMO treatment (without further treatment after immunotherapy)

  - Missing data
  - Overlapping propensity score range

- Propensity score and exact matching (3:1)

  - **Complete-case control patients**
  - **Not missing any key covariate data**

Figure 25. FDA Analysis of Patient Selection Process for Final Matched Populations in Externally Controlled Trial



HRNB = high-risk neuroblastoma; PS = propensity score; CR = complete response; VGPR = very good partial response; PR = partial response; ASCT = autologous stem cell transplantation; BM = bone marrow; EFS = event-free survival

Source: FDA Analysis

[Reference] https://www.accessdata.fda.gov/drugsatfda_docs/nda/2024/215500Orig1s000MultidisciplineR.pdf

Propensity score and exact matching

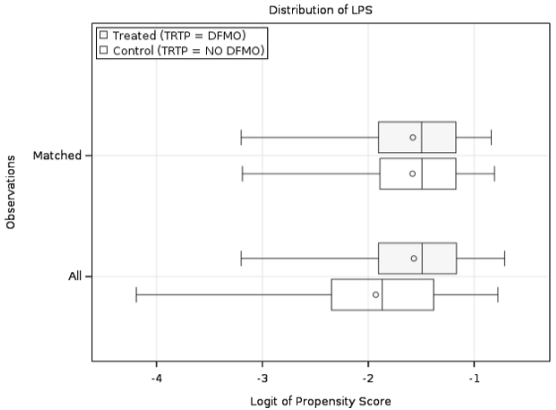
- 10 covariates in the propensity score included
  1. Age at high-risk diagnosis
  2. Sex
  3. Race
  4. Stage at diagnosis
  5. Pre-ASCT response
  6. Transplant type (single vs. tandem)
  7. Time from transplant to start of immunotherapy
  8. Response at end of immunotherapy
  9. Time from start of immunotherapy to end of immunotherapy
  10. Time from diagnosis to end of immunotherapy

[Reference] https://www.accessdata.fda.gov/drugsatfda_docs/nda/2024/215500Orig1s000MultidisciplineR.pdf

Propensity score and **exact matching**

- **Exact match:** MYCN status
  - The most important predictor of relapse and survival outcomes

Figure 28. Box plot of Logit Propensity Scores Before and After Matching



Source: FDA Analysis of Applicant submitted data 3bvanbl0032 ADTTE

Disease Characteristics

The Applicant's Position:

Similar to the demographic characteristics, disease characteristics are comparable between overall treated and control groups, though again there is some missing data. Good balance across all disease characteristics is observed in the matched cohorts of treated and control patients.

Table 25: Disease and Prior Treatment Characteristics DFMO PER COG and NO DFMO Population

	NO DFMO			DFMO PER COG		
	(N=852) n (%)	Complete Case[1] (N=516) n (%)	Matched[2] (N=270) n (%)	(N=92) n (%)	Complete Case[1] (N=91) n (%)	Matched[2] (N=90) n (%)
Age at High-risk Diagnosis (Years)						
0 - < 18 months	93 (10.9)	79 (15.3)	46 (17.0)	12 (13.0)	12 (13.2)	12 (13.3)
>= 18 months - < 6 years	497 (58.3)	366 (70.9)	190 (70.4)	68 (73.9)	68 (74.7)	67 (74.4)
>= 6 years	98 (11.5)	71 (13.8)	34 (12.6)	12 (13.0)	11 (12.1)	11 (12.2)
Unknown	164 (19.2)	0	0	0	0	0
n, Mean (SD)	688, 3.7 (2.49)	516, 3.6 (2.56)	270, 3.6 (2.59)	92, 3.8 (2.86)	91, 3.7 (2.86)	90, 3.7 (2.87)
Stage at Diagnosis						
Stage 4	565 (66.3)	414 (80.2)	233 (86.3)	80 (87.0)	79 (86.8)	78 (86.7)
Other	123 (14.4)	102 (19.8)	37 (13.7)	12 (13.0)	12 (13.2)	12 (13.3)
Unknown	164 (19.2)	0	0	0	0	0
MYCN						
Amplified	250 (29.3)	220 (42.6)	120 (44.4)	40 (43.5)	40 (44.0)	40 (44.4)
Not amplified	345 (40.5)	296 (57.4)	150 (55.6)	51 (55.4)	51 (56.0)	50 (55.6)
Unknown	257 (30.2)	0	0	1 (1.1)	0	0

[Reference] https://www.accessdata.fda.gov/drugsatfda_docs/nda/2024/215500Orig1s000MultidisciplineR.pdf

FDA analysis: **Confounding Factors**

- Social Determinants of Health
- Primary Tumor Location
- Cytogenetics

Table 45. FDA Analysis of Impact of Potential Unmeasured Confounding Factors on Treatment Effect Estimates

Potential Unmeasured Confounder		Supporting Literature	Association with Outcome	Prevalence in DFMO Arm	Prevalence in Control Arm	Adjusted DFMO HR (95% CI)
Applicant's Proposed Primary Analysis of EFS						0.48 (0.27, 0.85)
Social Determinants of Health	Household poverty	Bona (2021)	EFS HR = 1.9	35%	70%	0.59 (0.53, 0.67)
	Neighborhood & household poverty	Bona (2021)	EFS HR = 2.2	14%	42%*	0.62 (0.54, 0.71)
Primary Tumor Location	Adrenal vs. Non-adrenal	Vo (2014)	EFS HR = 1.1	47%	94%	0.50 (0.48, 0.52)
	Non-thoracic vs. Thoracic	Vo (2014)	EFS HR = 1.3	85%	100%**	0.50 (0.49, 0.50)
Applicant's Proposed Primary Analysis of OS						0.32 (0.15, 0.70)
Cytogenetics	Chromosome 1p deletion	Bown (1999)	OS HR = 1.9	47%	94%	0.42 (0.33, 0.48)
	Chromosome 17q gain	Bown (1999)	OS HR = 3.4	54%	100%**	0.47 (0.39, 0.52)

*Triple prevalence considered due to low expected prevalence; **If double prevalence exceeds 100%, the prevalence is capped to 100%

Source: FDA analysis

[Reference] https://www.accessdata.fda.gov/drugsatfda_docs/nda/2024/215500Orig1s000MultidisciplineR.pdf

FDA analysis: **Bias**

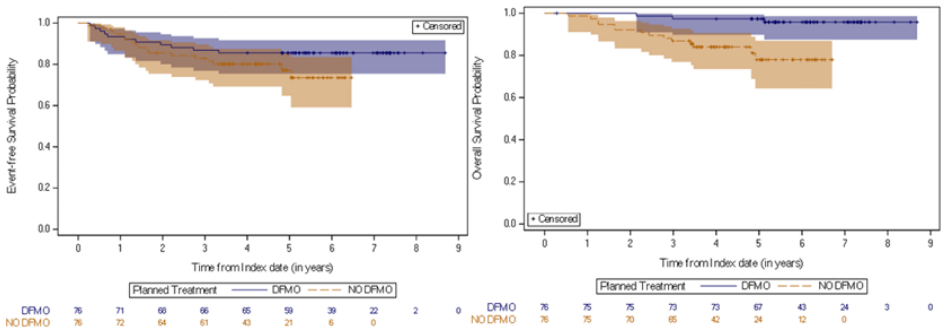
- 1:1 Matching

Table 46: Sensitivity Analysis of EFS and OS with Simultaneous Adjustment for Multiple Potential Sources of Bias– Propensity Score Matched (1:1)

	Event-free Survival			Overall Survival		
	DFMO Events/N	NO DFMO Events/N	HR (95% CI)	DFMO Events/N	NO DFMO Events/N	HR (95% CI)
Primary Analysis	14/90	79/270	0.48 (0.27, 0.85)	7/90	57/270	0.32 (0.15, 0.70)
Analysis with simultaneous exclusions/adjustments*	11/76	17/76	0.59 (0.28, 1.27)	3/76	14/76	0.16 (0.05, 0.57)

HR, hazard ratio; CI, confidence interval; *1:1 match due to reduced sample size
Source: FDA Analysis based on Applicant submitted data: 3bvanbl0032 ADTTE

Figure 39. Kaplan-Meier Plots of EFS (left) and OS (right) in the Sensitivity Analysis With Simultaneous Adjustment for Multiple Potential Sources of Bias- Propensity Score Matched (1:1)



[Reference] https://www.accessdata.fda.gov/drugsatfda_docs/nda/2024/215500Orig1s000MultidisciplineR.pdf

FDA: IWILFIN (eflornithine)

- **1st oncology approval** relying on an **externally controlled trial** as primary clinical data to support substantial evidence of effectiveness

ODAC Discussion Summary

Given the primary evidence of efficacy was from an ECT and available confirmatory data was from non-clinical studies, an ODAC was held in 2023 to discuss eflornithine (DFMO)

Uncertainties

- Potential biases due to non-randomized study design
- Limitations of animal models

Strengths

- Large effect size
- Robust statistical analyses
- Strong preclinical models

Discussion: highlighted challenges of application, feasibility of RCT

Committee voted (16 Yes – 4 No) that there was sufficient evidence to establish efficacy

[Reference] https://www.ema.europa.eu/en/documents/presentation/presentation-externally-controlled-trials-oncology-drivera-us-fda_en.pdf

Regulatory Insights

DFMO Regulatory Context

- 1st oncology approval relying on an **externally controlled trial** as primary clinical data to support substantial evidence of effectiveness

Clinical Context

- Rare, life-threatening cancer with high unmet need
- Feasibility concerns regarding successfully conducting new RCT

Evidentiary package unique and each component was essential for approval

- High-quality external control arm in a comparable population (matched on >10 clinical characteristics)
- Consistent estimation of the treatment effect through numerous sensitivity and supportive analyses
- Manageable safety profile for oncology product
- Strong nonclinical mechanistic data

- **RCTs remain strongly preferred** development approach to demonstrate effect on a time-to-event endpoint, even in rare cancers

23

[Reference] https://www.ema.europa.eu/en/documents/presentation/presentation-externally-controlled-trials-oncology-drivera-us-fda_en.pdf

Summary

- 현재 치료제가 없는 희귀질환과 같은 아주 제한적인 경우에만 RWE로 FDA의 신약 허가를 받을 수 있다
- FDA로부터 신약 허가를 받거나 적응증을 확장한 경우, RWD source는 eHR이나 레지스트리가 많았다
- 이 때 RWE 디자인은 비중재연구 (관찰연구) 또는 외부대조군 연구가 많았다
- 외부 대조군은 임상시험의 바깥에 있는 연구이며, 치료약을 투여 받지 않은 참여자로 구성된다
- 외부 대조군 연구는 자연사 연구로부터 기인하는 경우가 많으며, historical control로 사용된다
- 외부 대조군의 데이터는 RWD를 사용하거나, 다른 임상시험의 patient-level data를 사용할 수도 있다
- 최근 외부 대조군을 1차 유효성 평가 변수의 대조군으로 사용한 사례가 있다
- 해당 사례는 2023-12-13 일자로 FDA의 승인을 받은 IWILFIN (eflornithine) 이라는 약물로, 2023-10-04 일자로 Oncologic Drugs Advisory Committee (ODAC)의 실시간 투표를 거쳤다
- 성향점수맞춤 (Propensity score matching) 방법을 사용하여 비교한 결과, 효과가 우수함을 입증하였다
- 앞으로 RWD/RWE를 적용한 신약개발 사례는 더 많아질 것으로 기대한다

End of Document


RESEARCH
C&R
RESEARCH